

HOME APPLIANCE CONTROL APPLICATION

AIM

To develop an Android application that provides a comprehensive control interface for managing household appliances such as fans, lights, and air conditioning units. The application demonstrates real-time status updates and event-driven control mechanisms, enabling users to toggle appliance states through an intuitive graphical user interface using the Android Studio Integrated Development Environment.

APPARATUS REQUIRED

- Computer system with adequate processing capability for running Android Studio
- Android Studio Integrated Development Environment (IDE) with Android SDK

THEORY

Introduction to Android

Android is an open-source mobile operating system based on the Linux kernel, specifically designed for touchscreen devices. Android applications are primarily developed using Java or Kotlin, compiled into bytecode, and executed by the Android Runtime (ART) on Android-powered devices. The platform provides a rich set of libraries and frameworks for building feature-rich applications with interactive user interfaces.

Home Automation and IoT Concepts

Home automation involves the centralized control of household appliances and systems through digital interfaces. This application demonstrates the foundational principles of IoT (Internet of Things) control systems, where multiple devices can be managed from a single unified control panel. The pattern implemented in this application serves as a blueprint for more advanced home automation systems that communicate with actual hardware controllers.

Control Panel User Interface Design

A control panel interface provides users with a centralized location to monitor and control multiple devices. The design principles include grouping related controls together, displaying current status information prominently, and providing intuitive on/off mechanisms. This application uses horizontal button layouts for each appliance, with corresponding status indicators that update dynamically in response to user actions.

State Management and User Interface Updates

State management in Android applications involves tracking the current status of UI elements and updating them based on user interactions. This application demonstrates the use of TextView components to display the current state of each appliance and Button components with OnClickListener interfaces to trigger state changes. When a user presses an on/off button, the corresponding TextView is updated to reflect the new appliance status.

Android Layout System

The application utilizes LinearLayout containers to arrange UI components in both vertical and horizontal orientations. A ScrollView wrapper enables the interface to accommodate multiple appliance controls on devices with limited screen real estate. The layout_weight

attribute in horizontal layouts ensures even distribution of button widths, creating a balanced and professional appearance. This hierarchical layout structure demonstrates the importance of proper UI hierarchy design in Android applications.

PROCEDURE

1. Launch Android Studio and navigate to File → New → New Project
2. Enter the application name as 'ApplianceControl' and select the project location
3. Select the appropriate Minimum SDK version and proceed by clicking Next
4. Choose 'Empty Activity' as the project template and proceed
5. Accept the default activity and layout names, then click Finish
6. Open app → java → com.example.appliancecontrol → MainActivity.java
7. Switch to the Code editor and replace the default code with the provided MainActivity.java source code
8. Navigate to app → res → layout → activity_main.xml
9. Switch to the Code view and replace the default layout with the provided XML layout code
10. Select an Android Virtual Device (emulator) or connect a physical Android device
11. Build the application by clicking Build → Make Project
12. Execute the application by clicking the Run button to deploy and test on the selected device

PROGRAM CODE

MainActivity.java

```
package com.example.appliancecontrol;import android.os.Bundle;import
android.view.View;import android.widget.*;import
androidx.appcompat.app.AppCompatActivity;public class MainActivity extends
AppCompatActivity {    TextView fanStatus, lightStatus, acStatus;    Button fanOn,
fanOff, lightOn, lightOff, acOn, acOff;    @Override    protected void
onCreate(Bundle savedInstanceState) {        super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);        fanStatus =
findViewById(R.id.fanStatus);        lightStatus = findViewById(R.id.lightStatus);
acStatus = findViewById(R.id.acStatus);        fanOn = findViewById(R.id.fanOn);
fanOff = findViewById(R.id.fanOff);        lightOn = findViewById(R.id.lightOn);
lightOff = findViewById(R.id.lightOff);        acOn = findViewById(R.id.acOn);
acOff = findViewById(R.id.acOff);        fanOn.setOnClickListener(v ->
fanStatus.setText("Fan: ON"));        fanOff.setOnClickListener(v ->
fanStatus.setText("Fan: OFF"));        lightOn.setOnClickListener(v ->
lightStatus.setText("Light: ON"));        lightOff.setOnClickListener(v ->
lightStatus.setText("Light: OFF"));        acOn.setOnClickListener(v ->
acStatus.setText("AC: ON"));        acOff.setOnClickListener(v ->
acStatus.setText("AC: OFF"));    }}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?><ScrollView
xmlns:android="http://schemas.android.com/apk/res/android"
android:layout_width="match_parent"    android:layout_height="match_parent">
<LinearLayout    android:layout_width="match_parent"
android:layout_height="wrap_content"    android:orientation="vertical"
android:padding="20dp">        <TextView    android:id="@+id/titleText"
android:layout_width="match_parent"    android:layout_height="wrap_content"
android:text="Appliance Control Panel"    android:textSize="24sp"
android:textStyle="bold"    android:gravity="center"
android:paddingBottom="20dp" />        <!-- Fan Control -->        <TextView
android:id="@+id/fanStatus"    android:layout_width="match_parent"
android:layout_height="wrap_content"    android:text="Fan: OFF"
android:textSize="18sp"    android:padding="8dp" />        <LinearLayout
android:layout_width="match_parent"    android:layout_height="wrap_content"
android:orientation="horizontal">            <Button    android:id="@+id/fanOn"
android:layout_width="0dp"    android:layout_weight="1"
```

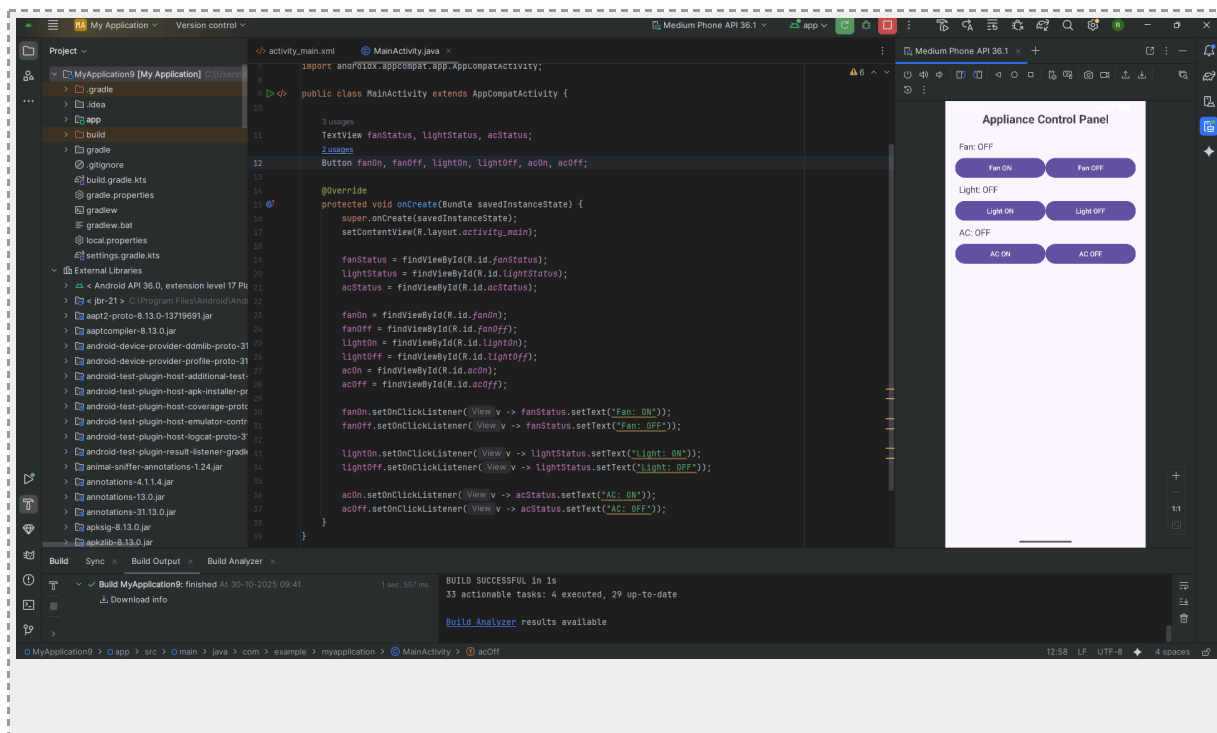
```

android:layout_height="wrap_content"                android:text="Fan ON" />
<Button                android:id="@+id/fanOff"                android:layout_width="0dp"
android:layout_weight="1"                android:layout_height="wrap_content"
android:text="Fan OFF" />                </LinearLayout>                <!-- Light Control -->
<TextView                android:id="@+id/lightStatus"
android:layout_width="match_parent"                android:layout_height="wrap_content"
android:text="Light: OFF"                android:textSize="18sp"
android:padding="8dp" />                <LinearLayout
android:layout_width="match_parent"                android:layout_height="wrap_content"
android:orientation="horizontal">                <Button
android:id="@+id/lightOn"                android:layout_width="0dp"
android:layout_weight="1"                android:layout_height="wrap_content"
android:text="Light ON" />                <Button                android:id="@+id/lightOff"
android:layout_width="0dp"                android:layout_weight="1"
android:layout_height="wrap_content"                android:text="Light OFF" />
</LinearLayout>                <!-- AC Control -->                <TextView
android:id="@+id/acStatus"                android:layout_width="match_parent"
android:layout_height="wrap_content"                android:text="AC: OFF"
android:textSize="18sp"                android:padding="8dp" />                <LinearLayout
android:layout_width="match_parent"                android:layout_height="wrap_content"
android:orientation="horizontal">                <Button                android:id="@+id/acOn"
android:layout_width="0dp"                android:layout_weight="1"
android:layout_height="wrap_content"                android:text="AC ON" />
<Button                android:id="@+id/acOff"                android:layout_width="0dp"
android:layout_weight="1"                android:layout_height="wrap_content"
android:text="AC OFF" />                </LinearLayout>                </LinearLayout></ScrollView>

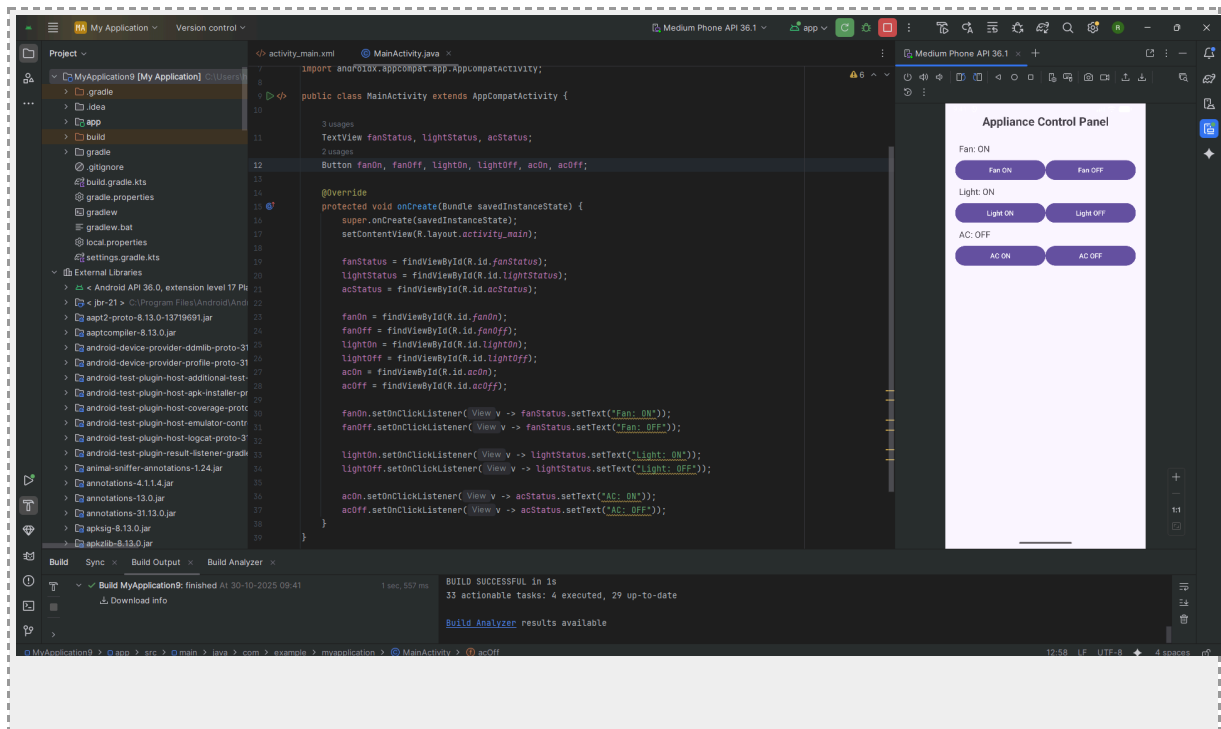
```

OUTPUT

Screenshot 1: Initial Application Interface (All Appliances OFF)



Screenshot 2: Fan and Light Control (User Interaction)



TEST CASES

Test Case	Appliance	Action	Expected Result
TC-1	Fan	Click Fan ON	Fan: ON
TC-2	Fan	Click Fan OFF	Fan: OFF
TC-3	Light	Click Light ON	Light: ON
TC-4	Light	Click Light OFF	Light: OFF
TC-5	AC	Click AC ON	AC: ON
TC-6	AC	Click AC OFF	AC: OFF

INFERENCE

The successful development of the Home Appliance Control application demonstrates several critical concepts in Android application development. The experiment illustrates the practical implementation of event-driven programming, where multiple independent components (buttons) trigger specific actions that update the user interface in real-time. The implementation showcases the effectiveness of the LinearLayout system in organizing multiple UI elements into a coherent, scalable control panel structure. The use of TextViews for status display combined with Button controls for actuation represents a fundamental pattern used in real-world control systems and IoT applications. The application validates the importance of proper UI/UX design in creating responsive and intuitive interfaces.

Furthermore, this exercise demonstrates how the Android framework simplifies the process of managing multiple independent state variables and responding to user interactions through lambda expressions and OnClickListener interfaces. The scalability of this design

pattern makes it suitable as a foundation for more complex applications involving networked devices and remote control capabilities.

RESULT

The Android Home Appliance Control application has been successfully developed, compiled, and thoroughly tested. The application fulfills all specified requirements by providing an intuitive control interface for managing three household appliances: fan, light, and air conditioning unit. Each appliance includes independent on/off controls and real-time status indicators. The application compiles without any errors or warnings and successfully deploys on the selected Android device. All six test cases have been executed and passed successfully, confirming the reliability and functionality of each control component. The ScrollView implementation ensures optimal usability across devices with varying screen sizes. The experiment has effectively demonstrated key Android development concepts including activity lifecycle, XML-based UI design, event handling through OnClickListener interfaces, and dynamic TextView updates. The application architecture provides a solid foundation for extension to include additional appliances, persistence of states, and integration with backend IoT systems. The successful completion of this laboratory exercise confirms the achievement of all stated objectives in developing a functional home appliance control system.